Chapter 11

2 : Consider the relation R = $\{(a,b),(a,c),(c,c),(b,b),(c,b),(b,c)\}$ on the set A = $\{a,b,c\}$. Is R reflexive? Symmetric? Transitive? If a property does not hold, say why.

$$\text{R is not reflexive} \because \{a,a\} \notin R$$

$$\text{R is not symmetric} \because \{a,b\} \in R \wedge \{b,a\} \notin R$$

$$\text{R is transitive} \because \begin{cases} \{a,b\},\{b,c\},\{a,c\} \in R \\ \{a,c\},\{c,b\},\{a,b\} \in R \\ \{c,b\} \in R \wedge \{b,a\} \notin R \\ \{b,c\} \in R \wedge \{c,a\} \notin R \end{cases}$$

12 : Prove that the relation | (divides) on the set $\mathbb{Z}$ is reflexive and transitive. (Use Example 11.8 as a guide if you are unsure of how to proceed.)

$$\mathbb{Z} \text{ is reflexive} \because x \mid x, \forall x \in \mathbb{Z}$$

$$\mathbb{Z} \text{ is transitive} \because \begin{cases} \text{if } x \mid y, \Rightarrow y = xk \text{ for some } k \in \mathbb{Z} \\ \text{if } y \mid z, \Rightarrow z = ym \text{ for some } k \in \mathbb{Z} \\ \text{therefore, } z = ym = xkm \Rightarrow x \mid z \end{cases}$$